

Introduction

Existing literature shows that urbanization affects and alters surface hydrology. Urban areas frequently experience reduced soil water percolation due to the compression of land by overlying infrastructure. As a result of such compression, only about 20% of water that falls on urban soil is absorbed while 80% flows through drainage systems and into rivers, contributing to flooding and soil dehydration (Markovi, G. et al., 2014). Impervious soil cover in cities also result in increased collection of anthropogenic pollutants by the impervious surfaces and their subsequent release into river systems during heavy rainfall events (Chithra S.V. et al., 2015) as well as increases in land surface temperatures and urban heat island effects (Yuan, Fei, and Marvin E. Bauer, 2007).

Loyola's Lakeshore Campus, located in an urban area, provides sustainable solutions to water conservation and stormwater diversion. This campus currently utilizes permeable pavement on major walkways, which helps to absorb stormwater and reduce flooding and runoff during heavy precipitation events.

Objectives and Future Goals

This multi-year project, which utilizes the field site set up in 2025 at the Loyola University Retreat and Ecology Campus (LUREC) by Dr. Debjani Ghatak and former undergraduate student Tristan Petersen (Fig 1), aims to quantify the efficiency of permeable surface cover compared to impermeable surface cover in influencing surface hydrology. The goal of this poster is to examine and model soil water content variability under different atmospheric conditions. Future goals include installation of temperature sensors and precipitation gauges at the LUREC site, covering the pots at the site with various cover materials (3 cement, 3 permeable tile, 3 native vegetation, and 3 uncovered), collecting on-site temperature and precipitation data, and statistically analyzing the data to examine differences in moisture content between cover materials and update current statistical models. The ultimate goal is to use these findings to inform modifications and improvements to Loyola's existing stormwater management systems.



Fig 1: The LUREC field site, which consists of a 3 by 4 array of cylindrical pots with Meter Group TEROS 10 soil moisture sensors installed in each pot. These sensors generate hourly readings of volumetric soil water content and transmit them via cables to ZL6 Basic data loggers. Each data logger receives readings from 6 of the 12 pots.

Methods

The research methodology includes (1) downloading volumetric water content readings collected by the sensors at the LUREC field site from February 4 to October 6, 2025, (2) analysis and visualization of data, and (3) generating a statistical model of daily changes in mean water content as a function of daily maximum temperature at a nearby weather station and the average of daily precipitation at three nearby weather stations. Weather station data was collected from NOAA's National Centers for Environmental Information website. Models were generated using Ordinary Least Squares multiple linear regression using the R statistical software.

Data visualization and modeling

Relationship Between Soil Water Content, Precipitation and Maximum Temperature Feb-Oct 2025

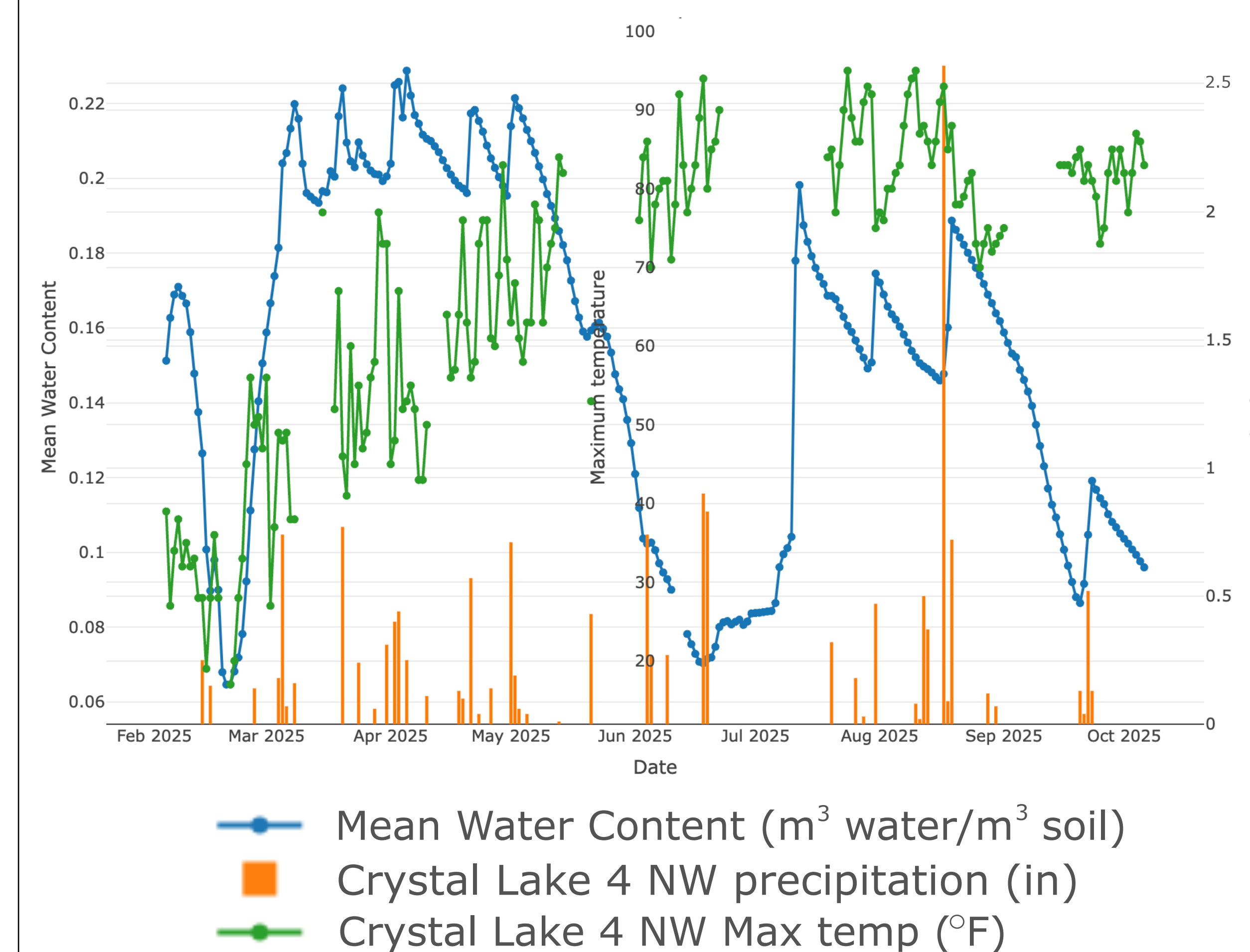


Fig 2: A time series graph of averaged mean water content for all 12 pots and Crystal Lake 4 NW weather station daily precipitation and maximum temperature. Shows less than one day of time delay between precipitation (inches) and increases in mean soil water content.

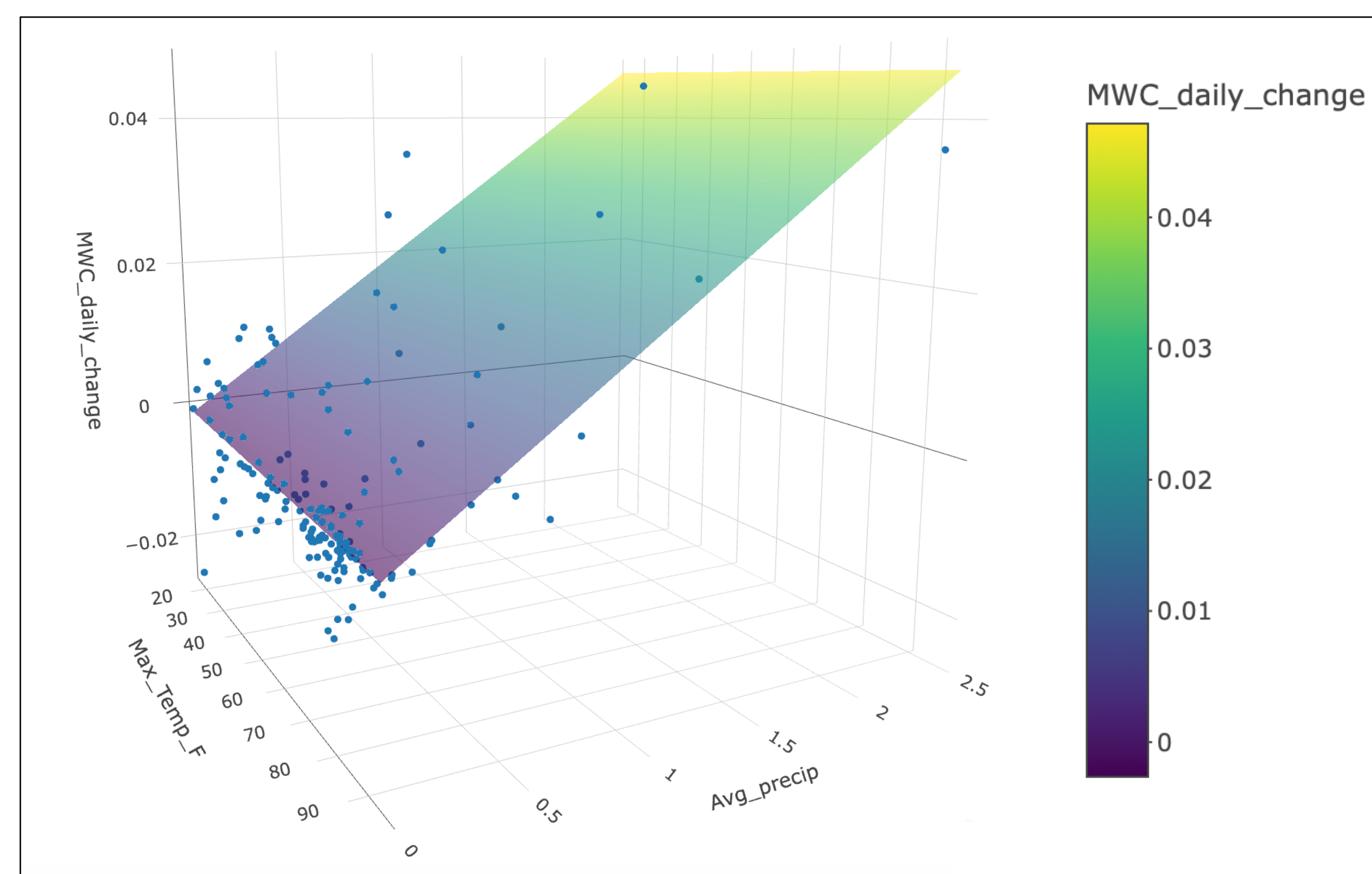


Fig 3: A multiple linear regression model of daily change in mean water content of all 12 pots (Z) vs Crystal Lake 4 NW maximum temperature (Y) and average daily precipitation of 3 nearby weather stations (X). The 3 stations used were Crystal Lake 4 NW, Woodstock 5 NW, and Woodstock 3.8 SW.

Model Results and Limitations

Given the current model, average precipitation is a statistically significant linear predictor of daily change in mean water content, but maximum temperature is not a significant predictor. The model contains two important limitations:

- Since average precipitation values are clustered heavily near zero, we need more observations to accurately examine the relationship between daily average precipitation and daily change in mean soil water content.
- Due to spatial variability of precipitation events and temperature, uncertainty arises from the use of proxies to estimate the daily precipitation and maximum temperature of the LUREC site.

These limitations could be overcome if temperature sensors and precipitation gauges were installed on site. Once soil cover materials are installed, new models can be generated from the onsite data to reassess the significance of temperature and precipitation in predicting mean water content changes, and to determine if these relationships differ across soil cover materials.

Conclusion

The analysis presented in this poster demonstrates the heavy role precipitation plays in regulating soil water content, as well as how daily changes in precipitation rapidly influence daily changes in soil water content. These results show that the water content sensors were installed successfully and continue to function efficiently. Since the LUREC field site pots are currently uncovered, the relationships examined in this study provide preliminary insight into the water infiltration of the control group (the uncovered pots), to be compared with the infiltration of pots with the cover materials and future, on-site temperature and precipitation measurements. Therefore, this poster highlights a crucial step towards generating useful insights for improving Loyola's existing stormwater diversion systems.

Acknowledgement

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References

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- Chithra, S. V., et al. "Impacts of impervious surfaces on the environment." *International Journal of Engineering Science Invention* 4.5 (2015): 27-31.
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External Data Sources

Ncei.Noaa.Gov. n.d. "Past Weather | National Centers for Environmental Information (NCEI)." Accessed April 4, 2026. <https://www.ncei.noaa.gov/access/past-weather/Woodstock%2C%20Illinois>.